

5. Find the domain and range of these functions. Note that in each case, to find the domain, determine the set of elements assigned values by the function.

a) the function that assigns to each bit string the number of ones in the string minus the number of zeros in the string.

Solution: Domain: The set of all possible bit strings.

~~Range:~~ Let, x be the number of 0's and y be the number of 1's of any arbitrary bit string S .

~~Range:~~ $(y-x)$ can be positive, negative and even 0.

$\therefore$ Range: $\mathbb{Z}$, $\because$ count is always an integer.

b) the function that assigns to each bit string twice the number of zeros in that string.

Solution: Domain: The set of all possible bit strings.

Let, x be the number of zeros of a bit string S .
 $2x$ must always be a natural number and is even

$\therefore$ Range: $\mathbb{E}$ The set of even natural no.s.

c) the function that assigns the number of bits left over when a bit string is split into bytes (which are blocks of 8 bits)

Solution: Domain: The set of all possible bit strings.

Range: $\{0, 1, 2, 3, 4, 5, 6, 7\}$, since the length of bit string S when divided by 8 gives these as remainders.

d) the function that assigns to each positive integer the largest perfect square not exceeding this integer.

Solution: Domain: $\mathbb{N} - \{0\}$

Range: The set of all positive perfect squares: $\{1, 4, 9, 16, 25, \dots\}$

7. Find the domain and range of these functions.

a) the function that assigns to each pair of positive integers the maximum of these two integers.

Solution: Domain: $(\mathbb{N} - \{0\}) \times (\mathbb{N} - \{0\})$ Range: $\mathbb{N} - \{0\}$